Southern University Bangladesh
Department of Computer Science and Engineering

TITLE OF THE LAB REPORT

Amplitude Modulation

Course Code: CSE 0611-232
Course Title: Data Communication Lab
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I. Objective:

- To generate and analyze an Amplitude Modulated (AM) signal using Python, and to observe its behavior in both time domain and frequency domain.

Program 1 (Simulation of Amplitude Modulation Signal**)**

Description:

In this program, a message signal and a carrier signal are generated using cosine functions in Python. The message signal has a low frequency, while the carrier signal has a much higher frequency.

Amplitude Modulation (AM) is performed by varying the amplitude of the carrier signal according to the message signal. The modulated signal is then plotted in the time domain to observe its waveform.

Additionally, the frequency components of the signals are analyzed using the Fast Fourier Transform. The frequency spectrum shows how the signal is distributed across different frequencies.

Algorithm:

- Import necessary libraries (NumPy, Matplotlib, FFT).
- Define parameters such as amplitude, frequency, modulation index, and sampling rate.
- Create a time vector using the sampling rate.
- Generate the message signal using a cosine function.
- Generate the carrier signal using a cosine function.
- Apply the AM modulation formula to generate the modulated signal:
  $s(t)=Ac(1+\mu \cdot m(t)Am)\cos(2\pi fct)$
- Plot the message, carrier, and modulated signals in the time domain.
- Use FFT to compute the frequency spectrum of each signal.
- Plot the frequency spectrum (only positive frequencies).
- Display total samples and frequency resolution.

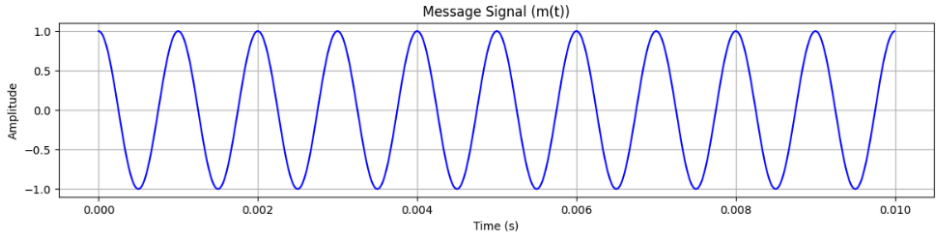
Source Code:

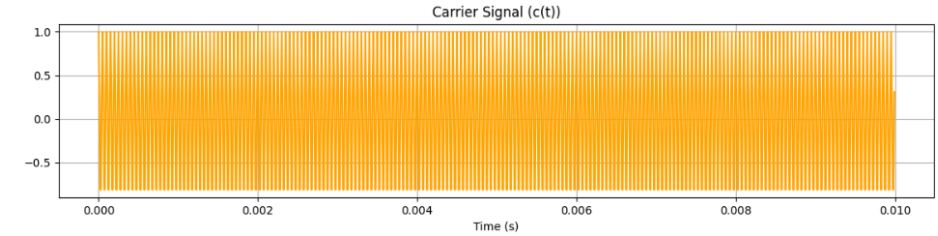
```
import numpy as np
import matplotlib.pyplot as plt
from scipy.fft import fft, fftfreq

Am = 1.0
fm = 1000
Ac = 1.0
fc = 20000
mu = 0.8
fs = 100000
duration = 0.01
```

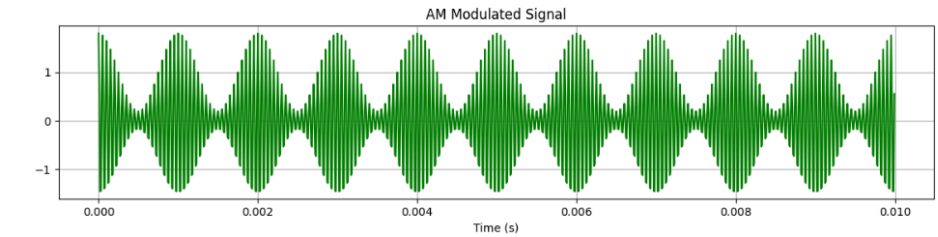
```
t = np.arange(0, duration, 1/fs)
message = Am * np.cos(2 * np.pi * fm * t)
carrier = Ac * np.cos(2 * np.pi * fc * t)


modulated = Ac * (1 + mu * (message / Am)) * np.cos(2 * np.pi * fc * t)

plt.figure(figsize=(12, 12))

plt.subplot(4, 1, 1)
plt.plot(t, message, color='blue')
plt.title('Message Signal (m(t))')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.grid(True)

plt.subplot(4, 1, 2)
plt.plot(t, carrier, color='orange')
plt.title('Carrier Signal (c(t))')
plt.xlabel('Time (s)')
plt.grid(True)

plt.subplot(4, 1, 3)
plt.plot(t, modulated, color='green')
plt.title('AM Modulated Signal')
plt.xlabel('Time (s)')
plt.grid(True)

plt.subplot(4, 1, 4)
plt.plot(t, modulated, color='green')
plt.xlim(0, 0.005)
plt.title('Zoomed AM Signal (Close look)')
plt.xlabel('Time (s)')
plt.grid(True)

plt.tight_layout()
plt.show()


def plot_spectrum(signal, title, fs_rate=fs):
```

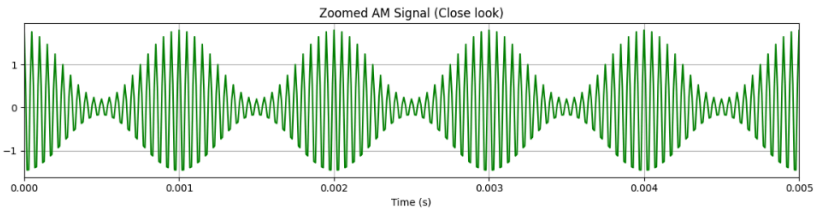
```
    N = len(signal)

    freq = fftfreq(N, 1/fs_rate)
    spectrum = np.abs(fft(signal)) / N

    plt.figure(figsize=(10, 4))

    plt.plot(freq[:N//2], 2 * spectrum[:N//2], color='red')
    plt.title(title, fontsize=12)
    plt.xlabel('Frequency (Hz)')
    plt.ylabel('Amplitude')
    plt.grid(True)

    plt.xlim(0, fc + 2*fm)
    plt.show()

print("\n--- Frequency Domain Analysis ---")
plot_spectrum(message, 'Frequency Spectrum of Message Signal')
plot_spectrum(carrier, 'Frequency Spectrum of Carrier Signal')
plot_spectrum(modulated, 'Frequency Spectrum of Modulated Signal')

print(f"Total Samples (N): {len(t)}")
print(f"Frequency Resolution: {fs/len(t)} Hz")
```

Program Output:

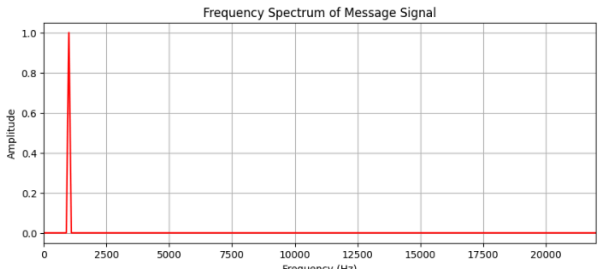


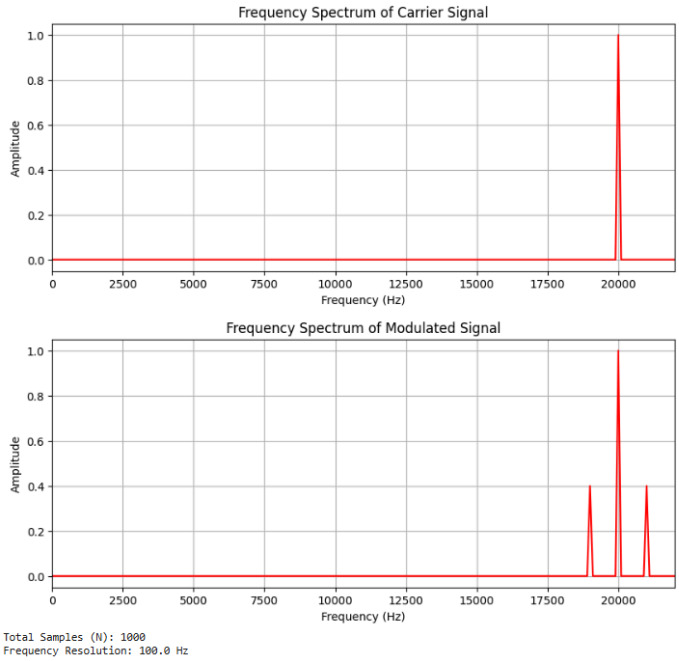







```
--- Frequency Domain Analysis ---
```





Discussion:

From this experiment, we observe that:

- The message signal contains only low-frequency components (1 kHz).
- The carrier signal has a high frequency (20 kHz).
- The AM modulated signal contains three main frequency components:
    - Carrier frequency (fc)
    - Upper sideband (fc+fm)
    - Lower sideband (fc−fm)

In the frequency spectrum, these components appear as distinct peaks.

Since the modulation index (μ=0.8) is less than 1, the signal is not over-modulated, and distortion is avoided.

This experiment helps in understanding:

- How amplitude modulation works
- The relationship between time domain and frequency domain
- The importance of AM in communication systems